aper code: CEN-505 Roll No. 22BCS074

B. Tech. (Computer Engineering) V Semester Examination, 2024-25

Time: 3 Hours **Computer Networks (Paper code: CEN-505)** **Max. Marks: 60**

Note: Attempt all questions. Make suitable assumption(s) wherever required.

Q1. (a)(i) For each OSI layer (Physical, Data Link, Network, and Transport), explain the process of encapsulation and decapsulation of data units as they move through the layers. [3] [BL2]

CO1

(ii) Discuss the purpose and functionality of addressing at each OSI layer (Data Link, Network, Transport and Application) and explain how addresses differ across the layers. [3] [BL2,4]

(b) An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (Maximum transmission Unit) is 100 bytes. Assume that the size of the IP header is 20 bytes. Calculate the number of fragments that the IP datagram will be divided into for transmission and determine the size of the last fragment. [6] [BL3,4]

OR

(b') An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter of it to Organization B, while retaining the remaining with itself. What are the possible range of addresses for organizations A and B? Also, calculate how many addresses are remained with the ISP? [6] [BL3,5]

Q2. (a) Describe how a packet-switched network can provide a connection-oriented service. In what ways does this method differ from the conventional circuit-switching approach? [6] [BL2,3]

CO2

(b) Illustrate how the choice of frequency band affects data transmission in unguided media. How does this influence the transmission method, such as sky-wave or line-of-sight? [6] [BL2,3]

OR

(b') Differentiate between bit rate and baud rate. What is the baud rate of Manchester encoding? Sketch the Manchester encoding and Differential Manchester encoding for the following bit stream: 10101011001. [6 [BL2,6

Q3. (a) Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial $x^3 + x + 1$. Suppose the message 01011011 is to be transmitted. Check bits are appended at the end of the message by the transmitter using the above CRC scheme. Determine the bit string to be transmitted. Also, show how this algorithm detects the error by making some errors in the transmitted message. [6 [BL3

CO3

(b)(i) Suppose two hosts are connected by a point-to-point link and they are configured to use Stop-and-Wait protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest. [3 [BL4

(a) Longer link length and lower transmission rate
(b) Longer link length and higher transmission rate
(c) Shorter link length and lower transmission rate
(d) Shorter link length and higher transmission rate
Justify your answer.

(ii) In a network using a Go-Back-3 (GB3) ARQ protocol, every 6th packet is lost. If you need to send 15 packets, how many retransmissions are required? [3] [BL3]

OR

(b') Consider a 128×10^3 bits/second satellite communication link with one way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgement. Calculate the minimum number of bits required for the sequence number field to achieve 100% utilization. [6] [BL5]

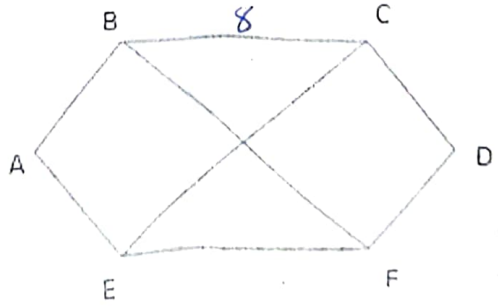
Q4. (a) How does virtual carrier sensing work in CSMA/CA, and why is it important for collision avoidance? What role does inter-frame spacing play in this process? [6 [BL2,3
CO4

(b) Discuss CSMA/CD. Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (108 bits per second) over a 1 km (kilometer) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes. What is the signal speed (km/sec) in the cable? [6 [BL2,5

OR

(b') Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1000 bits. The channel transmission rate is 1 Mbps. The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1000 frames per second. Calculate the throughput of the network (rounded to the nearest integer). [6 [BL5

Q5. (a) For the following figure, the network uses a Distance Vector Routing protocol. The following vectors have just come into router C: from B: (5,0,8,12,6,2); from D: (16,12,6,0,9,10); and from E: (7,6,3,9,0,4). The measured delays to B, D and E are 6, 3 and 5, respectively. What is C's new routing table? Give both the outgoing line to use and the expected delay [6 [BL5
CO5



ken bucket algorithm is used to control network traffic with a bucket size of 600 MB and oken arrival rate of 3 MB/sec. If a data burst of 1800 MB arrives at 30 MB/sec, and the aximum output rate is also 30 MB/sec, calculate the length of the burst at maximum rate and the total time needed to transmit the entire 1800 MB burst. [6] [BL5]

OR

What is the need of hierarchical routing? Consider a subnet with 720 routers. If there is no hierarchy, each router needs 720 routing entries. Determine the number of entries needed if three-level hierarchy is chosen, with eight clusters, each containing 9 regions of 10 routers. [6] [BL4]